

3 (a) pressure = force / area B1 [1]

(b) molecules collide with object / surface and rebound B1
 molecules have change in momentum hence force acts B1
 fewer molecules per unit volume on top of mountain / temperature is less B1
 hence lower speed of molecules A0
 hence less pressure [3]

(c) (i) $\rho = m / V$ C1
 $W = V\rho g = 0.25 \times 0.45 \times 9.81 \times 13600$ C1
 $= 15000 \text{ (15009) N}$ A1 [3]

(ii) $p = W / A$ (or using $p = \rho gh$) $= 15009 / 0.45$
 $= 3.3 \times 10^4 \text{ Pa}$ A1 [1]

(iii) pressure will be greater due to the air pressure (acting on the surface of the liquid)
 B1 [1]

4 (a) waves pass through the elements / gaps / slits in the grating M1